

4

Temperature and the Boltzmann factor

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In this chapter, we will explore the concept of *temperature* and show how it can be defined in a statistical manner. This leads to the idea of a *Boltzmann distribution* and a *Boltzmann factor*. Now of course the concept of **temperature** seems such an intuitively obvious one that you might wonder why we need a whole chapter to discuss it. Temperature is simply a measure of ‘hotness’ or ‘coldness’, so that we say that a hot body has a higher temperature than a cold one. For example, as shown in Fig. 4.1(a) if an object has temperature T_1 and is hotter than a second body with temperature T_2 , we expect that $T_1 > T_2$. But what do these numbers T_1 and T_2 signify? What does temperature actually mean?

4.1 Thermal equilibrium

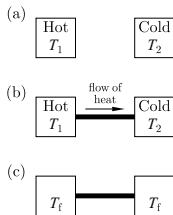


Fig. 4.1 (a) Two objects at different temperatures. (b) The objects are now placed in thermal contact and heat flows from the hot object to the cold object. (c) After a long time, the two objects have the same final temperature T_f .

To begin to answer these questions, let us consider what happens if our hot and cold bodies are placed in **thermal contact** which means that they are able to exchange energy. As described in Chapter 2, heat is ‘energy in transit’ and experiment suggests that, if nothing else is going on,¹ heat will always flow from the hotter body to the colder body, as shown in Fig. 4.1(b). This is backed up by our experience of the world: we always seem to burn ourselves when we touch something very hot (heat flows into us from the hot object) and become very chilled when we touch something very cold (heat flows out of us into the cold object). As heat flows from the hotter body to the colder body, we expect that the energy content and the temperatures of the two bodies will each change with time.

After some time being in thermal contact, we reach the situation in Fig. 4.1(c). The macroscopic properties of the two bodies are now no longer changing with time. If any energy flows from the first body to the second body, this is equal to the energy flowing from the second body to the first body; thus, there is *no net heat flow* between the two bodies. The two bodies are said to be in **thermal equilibrium**, which is defined by saying that the energy content and the temperatures of the

¹This is assuming that no additional power is being fed into the systems, such as occurs in the operation of a refrigerator which sucks heat out of the cold interior and dumps it into your warmer kitchen, but only because you are supplying electrical power.

two bodies will no longer be changing with time. We would expect that the two bodies in thermal equilibrium are now at the *same temperature*.

It seems that something irreversible has happened. Once the two bodies were put in thermal contact, the change from Fig. 4.1(b) to Fig. 4.1(c) proceeds inevitably. However, if we started with two bodies at the same temperature and placed them in thermal contact as in Fig. 4.1(c), the reverse process, i.e. ending up with Fig. 4.1(b), would not occur.² Thus as a function of time, systems in thermal contact tend towards thermal equilibrium, rather than away from it. The process that leads to thermal equilibrium is called **thermalization**.

If various bodies are all in thermal equilibrium with each other, then we would expect that their temperatures should be the same. This idea is encapsulated in the **zeroth law of thermodynamics**, which states that

Zeroth law of thermodynamics:
Two systems, each separately in thermal equilibrium with a third, are in equilibrium with each other.

You can tell by the numbering of the law that although it is an assumption that comes before the other laws of thermodynamics, it was added after the first three laws had been formulated. Early workers in thermodynamics took the content of the zeroth law as so obvious it hardly needed stating, and you might well agree with them! Nevertheless, the zeroth law gives us some justification for how to actually measure temperature: we place the body whose temperature needs to be measured in thermal contact with a second body which displays some property which has a well-known dependence on temperature and wait for them to come into thermal equilibrium. The second body is called a **thermometer**. The zeroth law then guarantees that if we have calibrated this second body against any other standard thermometer, we should always get consistent results. Thus, a more succinct statement of the zeroth law³ is: ‘thermometers work’.

²Thermal processes thus define an arrow of time. We will return to this point later in Section 34.5.

³This version is from our colleague M.G. Bowler.

4.2 Thermometers

We now make some remarks concerning thermometers.

- For a thermometer to work well, its heat capacity must be much lower than that of the object whose temperature one wants to measure. If this is not the case, the action of measurement (placing the thermometer in thermal contact with the object) could alter the temperature of the object.
- A common type of thermometer utilizes the fact that liquids expand when they are heated. Galileo Galilei used a water thermometer based on this principle in 1593, but it was Daniel Gabriel Fahrenheit (1686–1736) who devised thermometers based on alco-

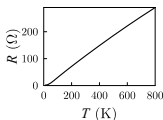


Fig. 4.2 The temperature dependence of the resistance of a typical platinum sensor.

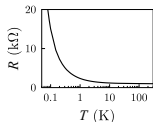


Fig. 4.3 The temperature dependence of the resistance of a typical RuO_2 sensor.

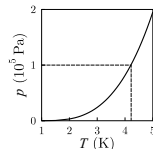


Fig. 4.4 The vapour pressure of ^4He as a function of temperature. The dashed line labels atmospheric pressure and the corresponding boiling point for liquid ^4He .

⁴We will introduce the Carnot engine in Section 13.2. The definition of temperature which arises from this is based upon eqn 13.7 and states that the ratio of the temperature of a body to the heat flow from it is a constant in a reversible Carnot cycle.

hol (1709) and mercury (1714) that bear most resemblance to modern household thermometers. He introduced his famous temperature scale which was then superseded by the more logical scheme devised by Anders Celsius (1701–1744).

- Another method is to measure the electrical resistance of a material which has a well-known dependence of resistance on temperature. Platinum is a popular choice since it is chemically resistant, ductile (so can be easily drawn into wires) and has a large temperature-coefficient of resistance; see Fig. 4.2. Other commonly used thermometers are based on doped germanium (a semiconductor which is very stable after repeated thermal cycling), carbon sensors and RuO_2 (in contrast to platinum, the electrical resistance of these thermometers increases as they are cooled; see Fig. 4.3).
- Using the ideal gas equation (eqn 1.12), one can measure the temperature of a gas by measuring its pressure with its volume fixed (or by measuring its volume with its pressure fixed). This works well as far as the ideal gas equation works, although at very low temperature, gases liquefy and show departures from the ideal gas equation.
- Another method which is useful in cryogenics is to have a liquid coexisting with its vapour and to measure the vapour pressure. For example, liquid helium (^4He , the most common isotope) has a vapour pressure dependence on temperature which is shown in Fig. 4.4.

All of these methods use some measurable property, like resistance or pressure, which depends in some, sometimes complicated, manner on temperature. However, none of them are completely linear across the entire temperature range of interest: mercury solidifies at very low temperature and becomes gaseous at very high temperature, the resistance of platinum saturates at very low temperature and platinum wire melts at very high temperature, etc. However, against what standard thermometer can one possibly assess the relative merits of these different thermometers? Which thermometer is perfect and gives the real thing, against which all other thermometers should be judged?

It is clear that we need some absolute definition of temperature based on fundamental physics. In the nineteenth century, one such definition was found, and it was based on a hypothetical machine, which has never been built, called a *Carnot engine*.⁴ Subsequently, it was found that temperature could be defined in terms of a purely statistical argument using ideas from probability theory, and this is the one we will use which we introduce in Section 4.4. In the following section we will introduce the terminology of *microstates* and *macrostates* that will be needed for this argument.

4.3 The microstates and macrostates

To make the distinction between microstates and macrostates, consider the following example.

Example 4.1

Imagine that you have a large box containing 100 identical coins. With the lid on the box, you give it a really good long and hard shake, so that you can hear the coins flipping, rattling and being generally tossed around. Now you open the lid and look inside the box. Some of the coins will be lying with heads facing up and some with tails facing up. There are lots of possible configurations that one could achieve (2^{100} to be precise, which is approximately 10^{30}) and we will assume that *each of these different configurations is equally likely*. Each possible configuration therefore has a probability of approximately 10^{-30} . We will call each particular configuration a **microstate** of this system. An example of one of these microstates would be: ‘Coin number 1 is heads, coin number 2 is heads, coin number 3 is tails, etc’. To identify a microstate, you would somehow need to identify each coin individually, which would be a bit of a bore. However, probably the way you would categorize the outcome of this experiment is by simply counting the number of coins which are heads and the number which are tails (e.g. 53 heads and 47 tails). This sort of categorisation we call a **macrostate** of this system. The macrostates are not equally likely. For example, of the $\approx 10^{30}$ possible individual configurations (microstates),

$$\begin{aligned} \# \text{ of configurations with 50 heads and 50 tails} &= \frac{100!}{(50!)^2} \approx 4 \times 10^{27}, \\ \# \text{ of configurations with 53 heads and 47 tails} &= \frac{100!}{53!47!} \approx 3 \times 10^{27}, \\ \# \text{ of configurations with 90 heads and 10 tails} &= \frac{100!}{90!10!} \approx 10^{13}, \\ \# \text{ of configurations with 100 heads and 0 tails} &= 1. \end{aligned}$$

Thus, the outcome with all 100 coins with heads facing up is a very unlikely outcome. This macrostate contains a single microstate. If that were the result of the experiment, you would probably conclude that (i) your shaking had not been very vigorous and that (ii) someone had carefully prepared the coins to be lying heads up at the start of the experiment. Of course, a *particular* microstate with 53 heads and 47 tails is just as unlikely; it is just that there are about 3×10^{27} other microstates which have 53 heads and 47 tails which look extremely similar.

This simple example shows two crucial points:

- The system could be described by a very large number of *equally likely* microstates.
- What you actually measure⁵ is a property of the macrostate of the

⁵In our example, the measurement was opening the large box and counting the number of coins which were heads and those which were tails.

system. The macrostates are *not* equally likely, because different macrostates correspond to different numbers of microstates.

The most likely macrostate that the system will find itself in is the one which corresponds to the largest number of microstates.

Thermal systems behave in a very similar way to the example we have just considered. To specify a microstate for a thermal system, you would need to give the microscopic configurations (perhaps position and velocity, or perhaps energy) of each and every atom in the system. In general it is impossible to measure which microstate the system is in. The macrostate of a thermal system on the other hand would be specified only by giving the macroscopic properties of the system, such as the pressure, the total energy or the volume. A macroscopic configuration, such as a gas with pressure 10^5 Pa in a volume 1 m^3 , would be associated with an enormous number of microstates. In the next section, we are going to give a statistical definition of temperature which is based on the idea that a thermal system can have a large number of equally likely microstates, but you are only able to measure the macrostate of the system. At this stage, we are *not* going to worry about what the microstates of the system actually are; we are simply going to posit their existence and say that if the system has energy E , then it could be in any one of $\Omega(E)$ equally likely microstates, where $\Omega(E)$ is some enormous number.

4.4 A statistical definition of temperature

We return to our example of Section 4.1 and consider two large systems which can exchange energy with each other, but not with anything else (Fig. 4.5). In other words, the two systems are in thermal contact with each other, but thermally isolated from their surroundings. The first system has energy E_1 and the second system has energy E_2 . The total energy $E = E_1 + E_2$ is therefore assumed fixed since the two systems cannot exchange energy with anything else. Hence the value of E_1 is enough to determine the macrostate of this joint system. Each of these systems can be in a number of possible microstates. This number of possible microstates could in principle be calculated as in Section 1.4 (and in particular, Example 1.3) and will be a very large, combinatorial number, but we will not worry about the details of this. Let us assume that the first system can be in any one of $\Omega_1(E_1)$ microstates and the second system can be in any one of $\Omega_2(E_2)$ microstates. Thus the whole system can be in any one of $\Omega_1(E_1)\Omega_2(E_2)$ microstates.⁶

The systems are able to exchange energy with each other, and we will assume that they have been left in the condition of being joined together for a sufficiently long time that they have come into *thermal equilibrium*. This means that E_1 and E_2 have come to fixed values. The crucial insight which we must make is that a *system will appear to choose a macroscopic configuration which maximizes the number of microstates*. This idea is based upon the following assumptions:

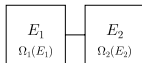


Fig. 4.5 Two systems able to exchange energy between them.

⁶We use the product of the two quantities, $\Omega_1(E_1)$ and $\Omega_2(E_2)$, because for each of the $\Omega_1(E_1)$ states of the first system, the second system can be in any of its $\Omega_2(E_2)$ different states. Hence the total number of possible combined states is the product of $\Omega_1(E_1)$ and $\Omega_2(E_2)$.

- (1) each one of the possible microstates of a system is equally likely to occur;
- (2) the system's internal dynamics are such that the microstates of the system are continually changing;
- (3) given enough time, the system will explore all possible microstates and spend an equal time in each of them.⁷

⁷This is the so-called ergodic hypothesis.

These assumptions imply that the system will most likely be found in a configuration which is represented by the most microstates. For a large system our phrase ‘most likely’ becomes ‘absolutely, overwhelmingly likely’; what appears at first sight to be a somewhat weak, probabilistic statement (perhaps on the same level as a five-day weather forecast) becomes an utterly reliable prediction on whose basis you can design an aircraft engine and trust your life to it!

For our problem of two connected systems, the most probable division of energy between the two systems is the one which maximizes $\Omega_1(E_1)\Omega_2(E_2)$, because this will correspond to the greatest number of possible microstates. Our systems are large and hence we can use calculus to study their properties; we can therefore consider making infinitesimal changes to the energy of one of the systems and seeing what happens. Therefore, we can maximize this expression with respect to E_1 by writing

$$\frac{d}{dE_1}(\Omega_1(E_1)\Omega_2(E_2)) = 0 \quad (4.1)$$

and hence, using standard rules for differentiation of a product, we have

$$\Omega_2(E_2)\frac{d\Omega_1(E_1)}{dE_1} + \Omega_1(E_1)\frac{d\Omega_2(E_2)}{dE_2}\frac{dE_2}{dE_1} = 0. \quad (4.2)$$

Since the total energy $E = E_1 + E_2$ is assumed fixed, this implies that

$$dE_1 = -dE_2, \quad (4.3)$$

and hence

$$\frac{dE_2}{dE_1} = -1, \quad (4.4)$$

so that eqn 4.2 becomes

$$\frac{1}{\Omega_1}\frac{d\Omega_1}{dE_1} - \frac{1}{\Omega_2}\frac{d\Omega_2}{dE_2} = 0, \quad (4.5)$$

and hence

$$\frac{d \ln \Omega_1}{dE_1} = \frac{d \ln \Omega_2}{dE_2}. \quad (4.6)$$

This condition defines the most likely division of energy between the two systems if they are allowed to exchange energy since it maximizes the total number of microstates. This division of energy is, of course, more usually called ‘being at the same temperature’, and so we identify $d \ln \Omega / dE$ with the temperature T (so that our two systems have $T_1 = T_2$). We will define the **temperature** T by

$$\frac{1}{k_B T} = \frac{d \ln \Omega}{dE}, \quad (4.7)$$

We will see later (Section 14.5) that in statistical mechanics, the quantity $k_B \ln \Omega$ is called the **entropy**, S , and hence eqn 4.7 is equivalent to

$$\frac{1}{T} = \frac{dS}{dE}.$$

where k_B is the **Boltzmann constant**, which is given by

$$k_B = 1.3807 \times 10^{-23} \text{ J K}^{-1}. \quad (4.8)$$

With this choice of constant, T has its usual interpretation and is measured in Kelvin. We will show in later chapters that this choice of definition leads to experimentally verifiable consequences, such as the correct expression for the pressure of a gas.

4.5 Ensembles

We are using probability to describe thermal systems and our approach is to imagine repeating an experiment to measure a property of a system again and again because we cannot control the microscopic properties (as described by the system's microstates). In an attempt to formalize this, Josiah Willard Gibbs in 1878 introduced a concept known as an **ensemble**. This is an idealization in which one considers making a large number of mental 'photocopies' of the system, each one of which represents a possible state the system could be in. There are three main ensembles that tend to be used in thermal physics:

- (1) The **microcanonical ensemble**: an ensemble of systems that each have the same fixed energy.
- (2) The **canonical ensemble**: an ensemble of systems, each of which can exchange its energy with a large reservoir of heat. As we shall see, this fixes (and defines) the temperature of the system.
- (3) The **grand canonical ensemble**: an ensemble of systems, each of which can exchange both energy and particles with a large reservoir. (This fixes the system's temperature and a quantity known as the system's chemical potential. We will not consider this again until Chapter 22 and it can be ignored for the present.)

In the next section we will consider the canonical ensemble in more detail and use it to derive the probability of a system at a fixed temperature being in a particular microstate.

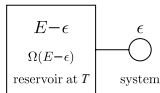


Fig. 4.6 A large reservoir (or heat bath) at temperature T connected to a small system.

4.6 Canonical ensemble

We now consider two systems coupled as before in such a way that they can exchange energy (Fig. 4.6). This time, we will make one of them enormous, and call it the **reservoir** (also known as a **heat bath**). It is so large that you can take quite a lot of energy out of it and yet it can remain at essentially the same temperature. In the same way, if you stand on the sea shore and take an eggcup-full of water out of the ocean, you do not notice the level of the ocean going down (although it does in fact go down, but by an unmeasurably small amount). The number of ways of arranging the quanta of energy of the reservoir will therefore be colossal. The other system is small and will be known as

the **system**. We will assume that for each allowed energy of the system there is only a single microstate, and therefore the system has a value of Ω equal to one. Once again, we fix⁸ the total energy of the system plus reservoir to be E . The energy of the reservoir is taken to be $E - \epsilon$ while the energy of the system is taken to be ϵ . This situation of a system in thermal contact with a large reservoir is very important and is known as the **canonical ensemble**.⁹

The probability $P(\epsilon)$ that the system has energy ϵ is proportional to the number of microstates which are accessible to the reservoir multiplied by the number of microstates which are accessible to the system. This is therefore

$$P(\epsilon) \propto \Omega(E - \epsilon) \times 1. \quad (4.9)$$

Since we have an expression for temperature in terms of the logarithm of Ω (eqn 4.7), and since $\epsilon \ll E$, we can perform a Taylor expansion¹⁰ of $\ln \Omega(E - \epsilon)$ around $\epsilon = 0$, so that

$$\ln \Omega(E - \epsilon) = \ln \Omega(E) - \frac{d \ln \Omega(E)}{dE} \epsilon + \dots \quad (4.10)$$

and so now using eqn. 4.7, we have

$$\ln \Omega(E - \epsilon) = \ln \Omega(E) - \frac{\epsilon}{k_B T} + \dots, \quad (4.11)$$

where T is the temperature of the reservoir. In fact, we can neglect the further terms in the Taylor expansion (see Exercise 4.4) and hence eqn 4.11 becomes

$$\Omega(E - \epsilon) = \Omega(E) e^{-\epsilon/k_B T}. \quad (4.12)$$

Using eqn 4.9 we thus arrive at the following result for the probability distribution describing the system which is given by

$$P(\epsilon) \propto e^{-\epsilon/k_B T}. \quad (4.13)$$

Since the system is now in equilibrium with the reservoir, it also must have the same temperature as the reservoir. But notice that although the system therefore has fixed temperature T , its energy ϵ is not a constant but is governed by the probability distribution in eqn 4.13 (and is plotted in Fig. 4.7). This is known as the **Boltzmann distribution** and also as the **canonical distribution**. The term $e^{-\epsilon/k_B T}$ is known as a **Boltzmann factor**.

We now have a probability distribution which describes exactly how a small system behaves when coupled to a large reservoir at temperature T . The system has a reasonable chance of achieving an energy ϵ which is less than $k_B T$, but the exponential in the Boltzmann distribution begins to quickly reduce the probability of achieving an energy much greater than $k_B T$. However, to quantify this properly we need to normalize the probability distribution. If a system is in contact with a reservoir and has a microstate r with energy E_r , then

$$P(\text{microstate } r) = \frac{e^{-E_r/k_B T}}{\sum_i e^{-E_i/k_B T}}, \quad (4.14)$$

⁸We thus treat the system plus reservoir as being in what is known as the **microcanonical ensemble**, which has fixed energy with each of its microstates being equally likely.

⁹'Canonical' means part of the 'canon', the store of generally accepted things one should know. It's an odd word, but we're stuck with it. Focussing on a system whose energy is *not* fixed, but which can exchange energy with a big reservoir, is something we do a lot in thermal physics and is therefore in some sense canonical.

¹⁰See Appendix B.

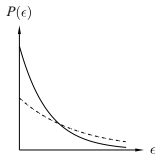


Fig. 4.7 The Boltzmann distribution. The dashed curve corresponds to a higher temperature than the solid curve.

The partition function is the subject of Chapter 20.

where the sum in the denominator makes sure that the probability is normalized. The sum in the denominator is called the **partition function** and is given the symbol Z .

We have derived the Boltzmann distribution on the basis of statistical arguments which show that this distribution of energy maximizes the number of microstates. It is instructive to verify this for a small system, so the following example presents the results of a computer experiment to demonstrate the validity of the Boltzmann distribution.

Example 4.2

To illustrate the statistical nature of the Boltzmann distribution, let us play a game in which quanta of energy are distributed in a lattice. We choose a lattice of 400 sites, arranged for convenience on a 20×20 grid. Each site initially contains a single energy quantum, as shown in Fig. 4.8(a). The adjacent histogram shows that there are 400 sites with one quantum on each. We now choose a site at random and remove the quantum from that site and place it on a second, randomly-chosen site. The resulting distribution is shown in Fig. 4.8(b), and the histogram shows that we now have 398 sites each with 1 quantum, 1 site with no quanta and 1 site with two quanta. This redistribution process is repeated many times and the resulting distribution is as shown in Fig. 4.8(c). The histogram describing this looks very much like a Boltzmann exponential distribution.

The initial distribution shown in Fig. 4.8(a) is very equitable and gives a distribution of energy quanta between sites of which Karl Marx would have been proud. It is however very *statistically unlikely* because it is associated with only a single microstate, i.e. $\Omega = 1$. There are many more microstates associated with other macrostates, as we shall now show. For example, the state obtained after a single iteration, such as the one shown in Fig. 4.8(b), is much more likely, since there are 400 ways to choose the site from which a quantum has been removed, and then 399 ways to choose the site to which a quantum is added; hence $\Omega = 400 \times 399 = 19600$ for this histogram (which contains 398 singly occupied sites, one site with zero quanta and one site with two quanta). The state obtained after many iterations in Fig. 4.8(c) is much, much more likely to occur if quanta are allowed to rearrange randomly as the number of microstates associated with the Boltzmann distribution is absolutely enormous. The Boltzmann distribution is simply a matter of probability.

In the model considered in this example, the rôle of temperature is played by the total number of energy quanta in play. So, for example, if instead the initial arrangement had been two quanta per site rather than one quantum per site, then after many iterations one would obtain the arrangement shown in Fig. 4.8(d). Since the initial arrangement has more energy, the final state is a Boltzmann distribution with a higher temperature (leading to more sites with more energy quanta).

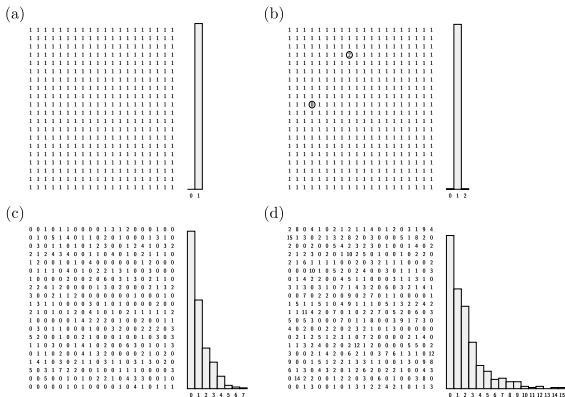


Fig. 4.8 Energy quanta distributed on a 20×20 lattice. (a) In the initial state, one quantum is placed on each site. (b) A site is chosen at random and a quantum is removed from that site and placed on a second randomly-chosen site. (c) After many repetitions of this process, the resulting distribution resembles a Boltzmann distribution. (d) The analogous final distribution following redistribution from an initial state with two quanta per site. The adjacent histogram in each case shows how many quanta are placed on each site.

Let us now start with a bigger lattice, now containing 10^6 sites, and place a quantum of energy on each site. We randomly move quanta from site to site as before, and in our computer program we let this proceed for a large number of iterations (in this case 10^{10}). The resulting distribution is shown in Fig. 4.9, which displays a graph on a logarithmic scale of the number of sites N with n quanta. The straight line is a fit to the expected Boltzmann distribution. This example is considered in more detail in the exercises.

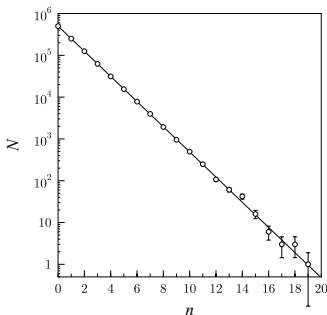


Fig. 4.9 The final distribution for a lattice of size 1000x1000 with one quantum of energy initially placed on each site. The error bars are calculated by assuming Poisson statistics and have length $\sqrt{N}$, where N is the number of sites having n quanta.

4.7 Applications of the Boltzmann distribution

To illustrate the application of the Boltzmann distribution, we now conclude this chapter with some examples. These examples involve little more than a simple application of the Boltzmann distribution, but they have important consequences.

Before we do so, let us introduce a piece of shorthand. Since we will often need to write the quantity $1/k_B T$, we will use the shorthand

$$\beta \equiv \frac{1}{k_B T}, \quad (4.15)$$

so that the Boltzmann factor becomes simply $e^{-\beta E}$. Using this shorthand, we can also write eqn 4.7 as

$$\beta = \frac{d \ln \Omega}{dE}. \quad (4.16)$$

Example 4.3

The two state system.

The first example is one of the simplest one can think of. In a two-state system, there are only two states, one with energy 0 and the other with energy $\epsilon > 0$. What is the average energy of the system?

Solution:

The probability of being in the lower state is given by eqn 4.14, so we have

$$P(0) = \frac{1}{1 + e^{-\beta\epsilon}}. \quad (4.17)$$

Similarly, the probability of being in the upper state is

$$P(\epsilon) = \frac{e^{-\beta\epsilon}}{1 + e^{-\beta\epsilon}}. \quad (4.18)$$

The average energy $\langle E \rangle$ of the system is then

$$\begin{aligned} \langle E \rangle &= 0 \cdot P(0) + \epsilon \cdot P(\epsilon) \\ &= \frac{\epsilon e^{-\beta\epsilon}}{1 + e^{-\beta\epsilon}} \\ &= \frac{\epsilon}{e^{\beta\epsilon} + 1}. \end{aligned} \quad (4.19)$$

This expression (plotted in Fig. 4.10) behaves as expected: when T is very low, $k_B T \ll \epsilon$, and so $\beta\epsilon \gg 1$ and $\langle E \rangle \rightarrow 0$ (the system is in the ground state). When T is very high, $k_B T \gg \epsilon$, and so $\beta\epsilon \ll 1$ and $\langle E \rangle \rightarrow \epsilon/2$ (both levels are equally occupied on average).

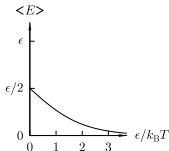


Fig. 4.10 The value of $\langle E \rangle$ as a function of $\epsilon/k_B T = \beta\epsilon$, following eqn 4.19. As $T \rightarrow \infty$, each energy level is equally likely to be occupied and so $\langle E \rangle = \epsilon/2$. When $T \rightarrow 0$, only the lower level is occupied and $\langle E \rangle = 0$.

Example 4.4

Isothermal atmosphere:

Estimate the number of molecules in an isothermal¹¹ atmosphere as a function of height.

Solution:

This is our first attempt at modelling the atmosphere, where we make the rather naive assumption that the temperature of the atmosphere is constant. Consider a molecule in an ideal gas at temperature T in the presence of gravity. The probability $P(z)$ of the molecule of mass m being at height z is given by

$$P(z) \propto e^{-mgz/k_B T}, \quad (4.20)$$

because its potential energy is mgz . Hence, the number density¹² of molecules $n(z)$ at height z , which will be proportional to the probability function $P(z)$ of finding a molecule at height z , is given by

$$n(z) = n(0)e^{-mgz/k_B T}. \quad (4.21)$$

This result (plotted in Fig. 4.11) agrees with a more pedestrian derivation which goes as follows: consider a layer of gas between height z and $z + dz$. There are $n dz$ molecules per unit area in this layer, and therefore they exert a pressure (force per unit area)

$$dp = -n dz \cdot mg \quad (4.22)$$

¹¹‘Isothermal’ means constant temperature. A more sophisticated treatment of the atmosphere is postponed until Section 12.4; see also Chapter 37.

¹²Number density means number per unit volume.

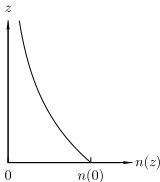


Fig. 4.11 The number density $n(z)$ of molecules at height z for an isothermal atmosphere.

downwards (because each molecule has weight mg). We note in passing that eqn 4.22 can be rearranged using $\rho = nm$ to show that

$$dp = -\rho g dz, \quad (4.23)$$

which is known as the **hydrostatic equation**. Using the ideal gas law (in the form derived in Chapter 6), which is $p = nk_B T$, we have that

$$\frac{dn}{n} = -\frac{mg}{k_B T} dz, \quad (4.24)$$

which is a simple differential equation yielding

$$\ln n(z) - \ln n(0) = -\frac{mg}{k_B T} z, \quad (4.25)$$

so that, again, we have

$$n(z) = n(0)e^{-mgz/k_B T}. \quad (4.26)$$

Our prediction is that the number density falls off exponentially with height, but the reality is different. Our assumption of constant T is at fault (the temperature falls as the altitude increases, at least initially) and we will return to this problem in Section 12.4, and also in Chapter 37.

Example 4.5

Chemical reactions:

Many chemical reactions have an activation energy E_{act} which is about $\frac{1}{2}$ eV. At $T = 300$ K, which is about room temperature, the probability that a particular reaction occurs is proportional to

$$\exp(-E_{\text{act}}/(k_B T)). \quad (4.27)$$

If the temperature is increased to $T + \Delta T = 310$ K, the probability increases to

$$\exp(-E_{\text{act}}/(k_B (T + \Delta T))), \quad (4.28)$$

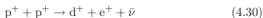
which is larger by a factor

$$\begin{aligned} \frac{\exp(-E_{\text{act}}/(k_B (T + \Delta T)))}{\exp(-E_{\text{act}}/(k_B T))} &= \exp\left(-\frac{E_{\text{act}}}{k_B}[(T + \Delta T)^{-1} - T^{-1}]\right) \\ &\approx \exp\left(\frac{E_{\text{act}}}{k_B T} \frac{\Delta T}{T}\right) \\ &\approx 2. \end{aligned} \quad (4.29)$$

Hence many chemical reactions roughly double in speed when the temperature is increased by about 10 degrees.

Example 4.6**The Sun:**

The main fusion reaction in the Sun¹³



but the main barrier to this occurring is the electrostatic repulsion of the two protons coming together in the first place. This energy is

$$E = \frac{e^2}{4\pi\epsilon_0 r}, \quad (4.31)$$

which for $r = 10^{-15}$ m, the distance which they must approach each other, E is about 1 MeV. The Boltzmann factor for this process at a temperature of $T \approx 10^7$ K (at the centre of the Sun) is

$$e^{-E/k_B T} \approx 10^{-400}. \quad (4.32)$$

This is extremely small, suggesting that the Sun is unlikely to undergo fusion. However, our lazy sunny afternoons are saved by the fact that quantum mechanical tunnelling allows the protons to pass *through* this barrier vastly more often than this calculation predicts that they could pass *over the top* of it.

¹³ p^+ is a proton, d^+ is a deuteron (a proton and a neutron), e^+ is a positron and $\bar{\nu}$ is a neutrino. This reaction and its consequences are explored more fully in Section 35.2.

Chapter summary

- The temperature T of a system is given by

$$\beta \equiv \frac{1}{k_B T} = \frac{d \ln \Omega}{dE},$$

where k_B is the Boltzmann constant, E is its energy, and Ω is the number of microstates (i.e. the number of ways of arranging the quanta of energy in the system).

- The *microcanonical ensemble* is an idealized collection of systems which all have the same fixed energy.
- The *canonical ensemble* is an idealized collection of systems, each of which can exchange its energy with a large reservoir of heat.
- For the canonical ensemble, the probability that a particular system has energy ϵ is given by

$$P(\epsilon) \propto e^{-\beta\epsilon}$$

(Boltzmann distribution), and the factor $e^{-\beta\epsilon}$ is known as the *Boltzmann factor*. Its use has been illustrated for a number of physical situations.

Further reading

Methods of measuring temperature are described in Pobell (1996) and White and Meeson (2002).

Exercises

- (4.1) Check that the probability in eqn 4.14 is normalized, so that the sum of all possible probabilities is one.
- (4.2) For the two-state system described in Example 3.2, derive an expression for the variance of the energy.
- (4.3) A system comprises N states which can have energy 0 or Δ . Show that the number of ways $\Omega(E)$ of arranging the total system to have energy $E = r\Delta$ (where r is an integer) is given by
- (4.5) A visible photon with energy 2 eV is absorbed by a macroscopic body held at room temperature. By what factor does Ω for the macroscopic body change? Repeat the calculation for a photon which originated from an FM radio transmitter.
- (4.6) Figure 4.10 is a plot of $\langle E \rangle$ as a function of $\beta\epsilon$. Sketch $\langle E \rangle$ as a function of temperature T (measured in units of ϵ/k_B).
- (4.7) Find the average energy $\langle E \rangle$ for

$$\Omega(E) = \frac{N!}{r!(N-r)!}. \quad (4.33)$$

Now remove a small amount of energy $s\Delta$ from the system, where $s \ll r$. Show that

$$\Omega(E - \epsilon) \approx \Omega(E) \frac{r^s}{(N-r)^s}, \quad (4.34)$$

and hence show that the system has temperature T given by

$$\frac{1}{k_B T} = \frac{1}{\Delta} \ln \left(\frac{N-r}{r} \right). \quad (4.35)$$

Sketch $k_B T$ as a function of r from $r = 0$ to $r = N$ and explain the result.

- (4.4) In eqn 4.11, we neglected the next term in the Taylor expansion which is
- $$\frac{d^2 \ln \Omega}{dE^2} \epsilon^2. \quad (4.36)$$
- Show that this term equals
- $$-\frac{\epsilon^2}{k_B T^2} \frac{dT}{dE}, \quad (4.37)$$
- and explain why this can be neglected compared to the first two terms if the reservoir is large. (Hint: how much should the temperature of the reservoir change when you change its energy by of order ϵ ?)
- (4.8) Estimate $k_B T$ at room temperature, and convert this energy into electronvolts (eV). Using this result, answer the following:
- (a) Would you expect hydrogen atoms to be ionized at room temperature? (The binding energy of an electron in a hydrogen atom is 13.6 eV.)
- (b) Would you expect the rotational energy levels of diatomic molecules to be excited at room temperature? (It costs about 10^{-4} eV to promote such a system to an excited rotational energy level.)
- (4.9) Write a computer program to reproduce the results in Example 3.1. For the case of $N \gg 1$ sites with initially one quantum per site, show that after many iterations you would expect there to be $N(n)$ sites with n quanta, where

$$N(n) \approx 2^{-n} N, \quad (4.38)$$

and explain why this is a Boltzmann distribution. Generalize your results for $Q \gg 1$ quanta distributed on $N \gg 1$ sites.